

Package #158

switching to irradiated-sterilized acetate-membrane or wet-membrane dialyzers

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented nephrology problem: maintenance hemodialysis patients receive a prescribed dialysis dose (commonly indexed as single-pool Kt/V, with guideline targets reported around 1.2-1.4 per session in the literature) that depends directly on the sustained blood flow rate through the extracorporeal circuit and the dialysate flow rate across the dialyser. When the delivered blood flow rate falls below the prescribed value (prescriptions commonly in the 300-500 mL/min range in the literature, with some reports of inadequate dose at lower effective flows), the delivered dose falls short and session time must be extended. Documented contributing factors include vascular access recirculation, arterial and venous pressure limits of the circuit, dialyser mass-transfer area and fiber bundle geometry, dialysate flow proportioning limits, and needle gauge resistance. This description is qualitative problem-domain context from the nephrology and biomedical-engineering literature; no specific incidence rates are asserted here (the engine's own retrieval must find and custody the evidence).

Mechanism: hypersensitivity to the hemodialyzer membrane (n = 11, 27.5%)

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.